

Table 2. Variants in four newly statistically significant and constrained ($pLI \geq 0.5$) ASD risk genes (*BRSK2*, *ITSN1*, *FEZF2*, and *PAX5*) in published and SPARK trios and associated phenotypic information

Subject ID	SP0037695	SP0042217	08C79336	SP0007556	SP0025011	SP0016887	13400.p1	13704.p1	14637.p1	SP0037344	11074.p1	SP0016232	12858.p1
Cohort	SPARK	SPARK	ASC	SPARK	SPARK	SPARK	SSC	SSC	SSC	SPARK	SSC	SPARK	SSC
Gene	<i>BRSK2</i>	<i>BRSK2</i>	<i>BRSK2</i>	<i>ITSN1</i>	<i>ITSN1</i>	<i>ITSN1</i>	<i>ITSN1</i>	<i>ITSN1</i>	<i>ITSN1</i>	<i>FEZF2</i>	<i>FEZF2</i>	<i>PAX5</i>	<i>PAX5</i>
Variant	p.T547fs	c.951-1G>A	c.1365-1G>C	p.P1619L (MPC = 2.03)	p.P156fs	p.Q711X	c.1952 + 1del	p.E576*	p. P164Rfs*22	p.A397fs	p.R344C (MPC = 3.37)	p.E113V (MPC = 2.78)	p.A111fs
Inheritance	de novo	de novo	de novo	de novo	Inherited (paternal)	Inherited (paternal)	Inherited (maternal)	Inherited (paternal)	Inherited (paternal)	de novo	de novo	de novo	de novo
Confirmed by Sanger sequencing?	YES	YES	Not available	YES	YES	YES	Not available	Not available	Not available	YES	YES	YES	YES
Sex	Male	Male	Male	Male	Male	Female	Female	Male	Male	Male	Male	Male	Female
DSM diagnosis	ASD	ASD	ASD	Asperger's disorder	Asperger's disorder	ASD	Autism	PDD-NOS	Autism	ASD	Autism	ASD	Autism
Age at evaluation (years)	8	19	5.3	30	34	2	4.2	4.3	5.4	23	9	14	4
Medical concerns	None	Premature birth (24 weeks), vision/hearing problems (not specified)	Unknown	Obesity, vision/hearing problems (not specified)	None	None	Migraines	None	None	None	None	Neurological problems (not specified), sleep disorder	None
Seizures (TRUE/FALSE)	FALSE	TRUE	FALSE	FALSE	FALSE	FALSE	FALSE	FALSE	FALSE	FALSE	Febrile seizures only	FALSE	FALSE
Intellectual Disability (TRUE/FALSE), IQ (if known)	TRUE	TRUE, ≤ 25	Unknown	FALSE, ≥ 129	FALSE	FALSE	FALSE, FSIQ = 116	FALSE, FSIQ = 91	TRUE – mild, FSIQ = 63	FALSE	TRUE, FSIQ = 68	TRUE, 55–69	FALSE, FSIQ = 91
Language level (at age of evaluation)	Delayed, single words	Delayed, no words	Delayed	Fluent speech (sentences)	Fluent speech (sentences)	Delayed, single words	Delayed, phrase speech (ADOS mod. 2)	Delayed, phrase speech (ADOS mod. 2)	Delayed, phrase speech (ADOS mod. 2)	Unknown	Fluent speech (sentences; ADOS mod. 3)	Delayed, no words	Delayed, phrase speech (ADOS mod. 2)
Language regression (TRUE/FALSE)	FALSE	FALSE	Unknown	FALSE	Unknown	TRUE	FALSE	TRUE	FALSE	Unknown	TRUE	TRUE	TRUE
Co-morbid psychological diagnoses	Learning disorder, motor skills delay, speech articulation problems, feeding disorder	Motor skills delay, speech articulation problems, feeding disorder, encopresis, enuresis	None	Attention or behavior problems—not specified, mood or anxiety problems—not specified, feeding disorder	Attention or behavior problems—not specified, OCD	Feeding disorder	Anxiety	None	None	None	None	Learning disorder, feeding disorder	None
Early motor delay (TRUE/FALSE)	FALSE	TRUE	FALSE	FALSE	Unknown	FALSE	FALSE	FALSE	FALSE	Unknown	FALSE	FALSE	FALSE

All damaging variants in SPARK participants within these genes have been confirmed with Sanger sequencing. Damaging variants in *PAX5* and *FEZF2* in the SSC were previously validated.³ MPC scores are listed for missense mutations. All phenotypic information for SPARK participants was collected online